Basic Statistics

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Outline I

1. Introduction

2. Different Data Types
   - Qualitative/Quantitative Aspects
   - Originality Aspects
   - Identity Aspects

Chapter 1: Introduction

Introduction I

What is statistics?

- Statistics is the science of making numerical conjectures about puzzling question.
  - What causes resemblance between parents and children? How strong is the force?
  - Are new medical treatments effective?
  - How much increment is needed in advertisement expenditure to attain a specific sales value?
  - Who is going to win the next election? What will be the margin?
  - Why does the casino make a profit?

Definition

- Statistics is the science of understanding data/information and of making decisions in the face of **variability** and **uncertainty.**

Introduction III

- Uncertainty:
  - There are many situations that we encounter in science (or more generally in life) in which the outcome is uncertain.
  - In some cases the uncertainty is because the outcome in question is not determined yet
    - We may not know whether it will rain tomorrow
  - In some cases the uncertainty is because although the outcome has been determined already, we are not aware of it
    - We may not know whether we passed a particular exam

Introduction IV

- Variability:
  - It can arise for various reasons.
  - If one visits a Teak plantation and measures the diameter of the trees at breast height, the measurements will reflect the natural variation from one tree to another,
  - Variation due to
    - Type of trees
    - Age differences
    - Measurement error
  - The concepts and methods of Statistics enable the investigator to describe/explain variability and to plan research so as to take variability into account.

Another definition

- Statistic: A body of scientific methods (statistical methods) used for collection, compilation, analysis and interpretation of statistical data.
  - Statistical methods are used to analyse data so as to extract the maximum information and also to quantify the message from that information.
  - Experiment->Observation->Inference

Chapter 2: Different Data Types in Statistics

Table: Information on 15 new-born babies, collected from a hospital

Sl No	Sex	Eye Color	Mother's Education	Weight (in k.g.)	Height (in c.m.)	No. of Siblings
1	female	blue	UG	3.85	52.51	1
2	female	black	HS	3.01	51.76	0
3	male	blue	UG	2.54	49.19	1
4	female	grey	PG	2.68	49.98	0
5	male	blue	HS	3.81	55.97	2
6	male	grey	HS	3.35	54.63	1
7	female	blue	UG	2.80	46.43	1
8	female	grey	PG	2.75	50.68	0
9	male	grey	UG	3.86	54.06	0
10	male	black	UG	3.76	54.04	1
11	female	blue	PG	2.59	47.01	2
12	male	black	HS	2.68	52.14	1
13	male	grey	PG	3.62	51.77	0
14	female	black	UG	3.12	50.23	1
15	male	black	PG	2.88	47.91	1

Data Types: Qualitative/Quantitative II

- Attribute/Categorical (Non-numerical to start with)
  - Nominal:-
    - Name or label without any order
    - Sex, Religion
    - Matching
    - *character* in R
  - Ordinal:-
    - Name or label with an order
    - Level of satisfaction, Degree of pain
    - Matching and sorting
    - *factor* in R

Data Types: Qualitative/Quantitative III

- Variable (Numerical)
  - Discrete:-
    - Take distinct and isolated values
    - No. of children, No. of car
    - All arithmetic operations
    - *integer* in R
  - Continuous:-
    - Take any value in a range
    - Height or weight of a person
    - All arithmetic operations
    - *numeric* in R

Data Types: Originality Aspects

- Primary Data
  - Investigators/Enumerators go directly to the field of inquiry and through observation, interview or direct measurement collect the relevant data.
  - First-hand data
- Secondary Data
  - Data collected from secondary source (research organization, journal etc.), which collected data for their own purpose and later stored the data for future use
  - Second-hand data

Data Types: Identity Aspects

- Non-frequency Data
  - Identity of the individual is important and is kept in mind throughout the study
  - Example: Annual production of steel in India; only the production values without reference to the years have little significance
- Frequency Data
  - Identity of the individual is not important, and a particular value is important
  - Example: a particular individual height figure is important without reference to the individual concerned